

Next Feeding Session Predictor

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§2.2 Energy Model

Prepared milk is the direct proxy for caloric energy: 1 ml = 1 unit of energy. The baby's daily target and hourly drain rate follow directly from the 150 ml/kg/day guideline:

$$D = w \cdot r \quad \lambda = \frac{D}{24}$$

w = baby weight (kg), r = 150 ml/kg/day (default), λ = hourly drain rate (ml/h). For $w = 6.9$ kg: $D = 1\,035$ ml, $\lambda = 43.1$ ml/h.

The standard interval is the time for one preferred bottle of m_0 ml of prepared milk to be fully spent at rate λ :

$$SI = \frac{m_0}{\lambda}$$

90 ml water $\rightarrow m_0 = 100$ ml milk, $\lambda = 43.1$ ml/h $\rightarrow SI \approx 2\text{h } 19\text{m}$

§4.1 The Intake Function

The app tracks the total milk received in a rolling 24-hour window. The credit contributed by bottle i (volume m_i , given at time t_i) evaluated at time T is:

$$c_i(T) = \begin{cases} m_i & \text{if } T - t_i \leq 24 \text{ h} \\ \max(0, m_i - \lambda(T - t_i - 24)) & \text{if } T - t_i > 24 \text{ h} \end{cases}$$

The smoothed 24-hour intake at time T is the sum over all feeds logged before T :

$$I(T) = \sum_{i=1}^n c_i(T)$$

The decay rate past 24 h is exactly λ — the same rate at which the baby burns energy. A bottle's credit diminishes at exactly the speed its nutritional value is consumed. In steady-state feeding, $I(T)$ oscillates between D (just before each feed) and $D + m_0$ (just after).

§5.1 Predictor A — Adjusted Volume at Standard Time

Predictor A answers: if I feed at the standard time, how much should I give? It computes the volume that restores intake to the equilibrium peak $D + m_0$:

$$T_A = t_{\text{last}} + SI$$

$$V_A = (D + m_0) - I(T_A)$$

At equilibrium $I(T_A) = D$, so $V_A = m_0$ — exactly the standard bottle.

The parent is shown V_A converted to water ml and decides which bottle to use. If $V_A \leq 0$ the baby is still above the equilibrium ceiling — Predictor A is greyed out. If V_A exceeds the stomach capacity cap it is clamped (§6).

§5.2 Predictor B — Standard Volume at Adjusted Time

Predictor B answers: when should I give my preferred bottle? It finds the time T_B at which the 24-hour intake has decayed back to exactly the daily target D , so that adding m_0 restores intake to $D + m_0$:

$$I(T_B) = D$$

Feed when the 24-hour intake returns to the daily target.

T_B is found by binary search on $[t_{\text{last}}, t_{\text{last}} + T_{\text{max}}]$. At equilibrium $T_B = t_{\text{last}} + SI$. With a deficit, $T_B < t_{\text{last}} + SI$ (feed earlier); with a surplus, $T_B > t_{\text{last}} + SI$ (feed later).

§6 Stomach Capacity Constraint

The total undigested milk in the stomach at time T is modelled as the sum of unmetabolised portions of all recent feeds:

$$S(T) = \sum_{i=1}^n \max(0, m_i - \lambda(T - t_i))$$

The constraint is $S(T_B) + m_0 \leq m_{\text{cap}}$, where m_{cap} is one standard bottle size above the preferred size. This gives a minimum waiting time:

$$\Delta t_{\text{min}} = \max(0, \frac{m_{\text{last}} + m_0 - m_{\text{cap}}}{\lambda})$$

Example: $\Delta t_{\min} = \frac{100+100-135}{43.1} \approx 1\text{h } 30\text{m}$

If Predictor B's T_B falls before $t_{\text{last}} + \Delta t_{\min}$, it is clamped upward with a note: "Stomach not ready — baby is underfed but needs time to digest."